CS425: Computer Networks - Homework 1

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Question 1

Problem Statement: Write a program (in any language) that generates an n-bit frame for transmission from a k-bit data block D and a $(n-k+1)$ bit CRC pattern P. Compile and run the program with at least two sets of inputs to confirm that this program is generating CRC patterns correctly. Modify the program to: (a) Generate message $k = 10$, (b) Use $P = 110101$ for a 15-bit frame T, (c) Generate transmission errors, (d) Apply CRC to determine if frame is accepted/discarded.

Solution: The source code for the CRC generator and error simulator is attached as a separate file (`crc_impl.cpp`) along with this report. It handles both custom user input and the specific test case ($k = 10, P = 110101$) required by the prompt. To run the code:

1. Compile: `g++ crc_impl.cpp -o crc`
2. Execute: `./crc`

```
Provide custom input? (Y/N): N

--- AUTOMATIC MODE ---

--- SENDER SIDE ---
Generated Data (10 bits): 0011000010
Generator P:              110101
Appended Data:            001100001000000
Calculated CRC:           01101
Transmitted Frame (15 bits): 001100001001101
```

(a) Sample input and generated CRC

```
Provide custom input? (Y/N): Y
Enter Data Block (binary only): 1001011001
Enter CRC Pattern (binary only): 110101

--- SENDER SIDE ---
Original Data: 1001011001
CRC Pattern:   110101
Appended Data: 100101100100000
Computed CRC:  01110
Transmitted Frame: 100101100101110 (Length: 15)
```

(b) Custom input confirming validation logic

Figure 1: Program Output Verification

Question 2

Problem Statement: In the Go-back-N ARQ mechanism using k-bit sequence numbers, why is the window size limited to $2^k - 1$ and not 2^k?

Solution: In Go-Back-N, the sequence numbers range from 0 to $2^k - 1$. If the window size were 2^k, the sender could send a full set of frames (0 to $2^k - 1$). If all these frames are received but all ACKs are lost, the sender will time out and retransmit frame 0. However, the receiver's window has already slid forward and is now expecting the *next* frame 0 (from the next cycle). The receiver cannot distinguish between a retransmission of the old frame 0 and the first transmission of a new frame 0, leading to a protocol failure. Limiting the window to $2^k - 1$ ensures there is always at least one sequence number's gap to avoid this ambiguity.

Question 3

Problem Statement: What is the maximum window size that can be used in the Selective-Reject ARQ mechanism that uses k-bit sequence numbers? Explain your answer.

Solution: The maximum window size is 2^{k-1} (half the sequence space). Selective-Reject requires that the sender and receiver windows do not overlap even in the worst-case scenario (all ACKs lost). If the window size $> 2^{k-1}$, the receiver's window of acceptable sequence numbers could overlap with the previous window's sequence numbers after a wrap-around. By keeping the size at 2^{k-1}, we ensure that a retransmitted old frame and a new frame never share the same sequence number within the current active windows.

Question 4

Problem Statement: A channel has a data rate of 4 kbps and a propagation delay of 20 ms. For what range of frame sizes does stop-and-wait give an efficiency of at least 50%?

Solution: Efficiency (U) is given by: $U = \frac{t_{frame}}{t_{frame}+2t_{prop}}$

We want $U \geq 0.5$, which implies $t_{frame} \geq 2t_{prop}$. Given $t_{prop} = 20$ ms $= 0.02$ s, then $t_{frame} \geq 0.04$ s. Since $t_{frame} = \frac{L}{R}$, where $R = 4000$ bps: $\frac{L}{4000} \geq 0.04 \implies L \geq 160$ bits.

Answer: Frame size $L \geq 160$ bits.

Question 5

Problem Statement: Consider a frame consists of one character of 4 bits. Assume bit error probability $p = 10^{-3}$. (a) Prob. of no errors? (b) Prob. of at least one error? (c) With parity bit, prob. of undetected error?

Solution: $n = 4, p = 0.001$.

(a) $P(\text{no error}) = (1-p)^n = (0.999)^4 \approx \mathbf{0.9960}$

(b) $P(\text{at least one}) = 1 - P(\text{no error}) = 1 - 0.9960 = \mathbf{0.0040}$

(c) With parity ($n = 5$), error is undetected if even number of errors occur (2 or 4): $P = \binom{5}{2}p^2(1-p)^3 + \binom{5}{4}p^4(1-p) \approx 10(10^{-6})(0.997) + 5(10^{-12}) \approx \mathbf{9.97 \times 10^{-6}}$

Question 6

Problem Statement: For $P = 110011$ and $M = 11100011$, find the CRC.

Solution: Message M with $n - k$ (5) zeros appended: 1110001100000.

110011	1110001100000
	110011
	00101111
	110011
	0111000
	110011
	00101100
	110011
	0111110
	110011
	001101 → Remainder: **11010**

Transmitted Frame: **1110001111010**.

Question 7

Problem Statement: (a) Encode 10010011011 using $P(x) = x^4 + x + 1$. (b) Detect error 100010000000000. (c) Detect error 100110000000000.

Solution: (a) $P = 10011$, Data = 100100110110000.

10011	100100110110000
	10011
	000010110
	10011
	0010111
	10011
	0010000
	10011
	00011 → Remainder: **1100**

Frame: **100100110111100**.

(b) Error $E(x)$ divided by $P(x)$ yields a non-zero remainder. **Detected.**

(c) Error $E(x)$ is a multiple of $P(x)$ (remainder 0). **Not Detected.**